

Juan Antonio Monleón de la Lluvia

Nuclear Engineer - PhD Student

CONTACT

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EDUCATION

Double Master in Industrial Engineering and Nuclear Science and Technology

Polytechnic
University of Madrid

2021-2023

Bachelor in Industrial Technology Engineering

Polytechnic
University of Madrid

2017-2021

SOFTWARE

Nuclear Codes: MCNP, Serpent,
NJOY, ADVANTG

Programming: Python, Git, VS Code,
Copilot

Documentation: LaTeX, Microsoft
365

ABOUT ME

I am a passionate professional dedicated to the nuclear sector, driven by a strong interest in research and innovation. As a proactive person, I constantly seek out new learning opportunities to expand my knowledge and skills. I enjoy applying my programming expertise to solve complex problems and enhance both my daily work and personal projects.

LANGUAGES

Spanish Native language

English C2 Level

French B2 Level

AWARDS

Best Academic Report for Nuclear Master

Polytechnic University of Madrid
(2022-2023)

Best Master Thesis Award

SCK CEN, Belgium (2023)

Best Poster Award

Journées des Thèses of IRSN (2024) -
[View Poster](#)

WORK EXPERIENCE

PhD in Neutronics

IRSN, France

Nov 2023 - Present

- Researching uncertainty quantification in neutronic simulations for studies on nuclear vessel ageing under irradiation
- Performing sensitivity analysis and neutron flux calculations to evaluate reactor vessel integrity
- Applying data assimilation techniques and leveraging nuclear data to refine simulation models

Research Internship

SCK CEN, Belgium

Sep 2022 - Jun 2023

- Completed Master's thesis titled "**Reducing Spectrum-Driven Uncertainties with Variance Reduction Techniques**"
- Conducted neutronic and deep shielding analyses using MCNP and ADVANTG for the MYRRHA Project

Master Thesis Research

UPM, Spain

Jun 2023 - Oct 2023

- Developed "**Enhancing the EXFOR nuclear data library with Machine Learning Techniques**"
- Implemented outlier detection methods for proton-induced reactions

Master Internship

ENUSA Advanced Industries, Spain

Oct 2021 - Apr 2022

- Designed and developed a Python program for the analysis and inspection of fuel assemblies and spent fuel

PUBLICATIONS AND CONFERENCES

- "**Sensitivity Analysis and Uncertainty Quantification in PWR Irradiation Ageing like problems**" - ANS Winter Conference RPSD, Orlando Nov 2024
- SNA+MC Conference, Paris Oct 2024
- Annual Meeting of the SNE (Spanish Nuclear Society), Cordoba Oct 2024

TRAINING

- Master in Management and Direction of SMEs | EFEM 2024-2025
- NJOY Course - NEA, Paris Dec 2024
- Sensitivity Analysis Summer School (SAMO) | UNIPR, Parma Jun 2024
- Nuclear Competitions Summer School | ENEEP at STU, Bratislava Jul 2024
- Course in "Neutron Detection" | ENEEP at JSI, Ljubljana Dec 2023

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| - Serpent Bootcamp NEA, Paris | Nov 2023 |
| - Course in "Nuclear Data for Energy and Non-Energy Applications" GREAT PIONEER | Dec 2022 |
| - Course in "Safe and Secure Operation of Nuclear Installations" ENEEP STU & CTU | Jan 2022 |
| - Course in "Nuclear Waste Management" CIEMAT | 2022 |
| - Speaker at International Workshop "Machine Learning in Nuclear Science and Technology Applications" UPM, Madrid | May 2021 |
| - Various Certifications in Python and Machine Learning | 2020-2022 |